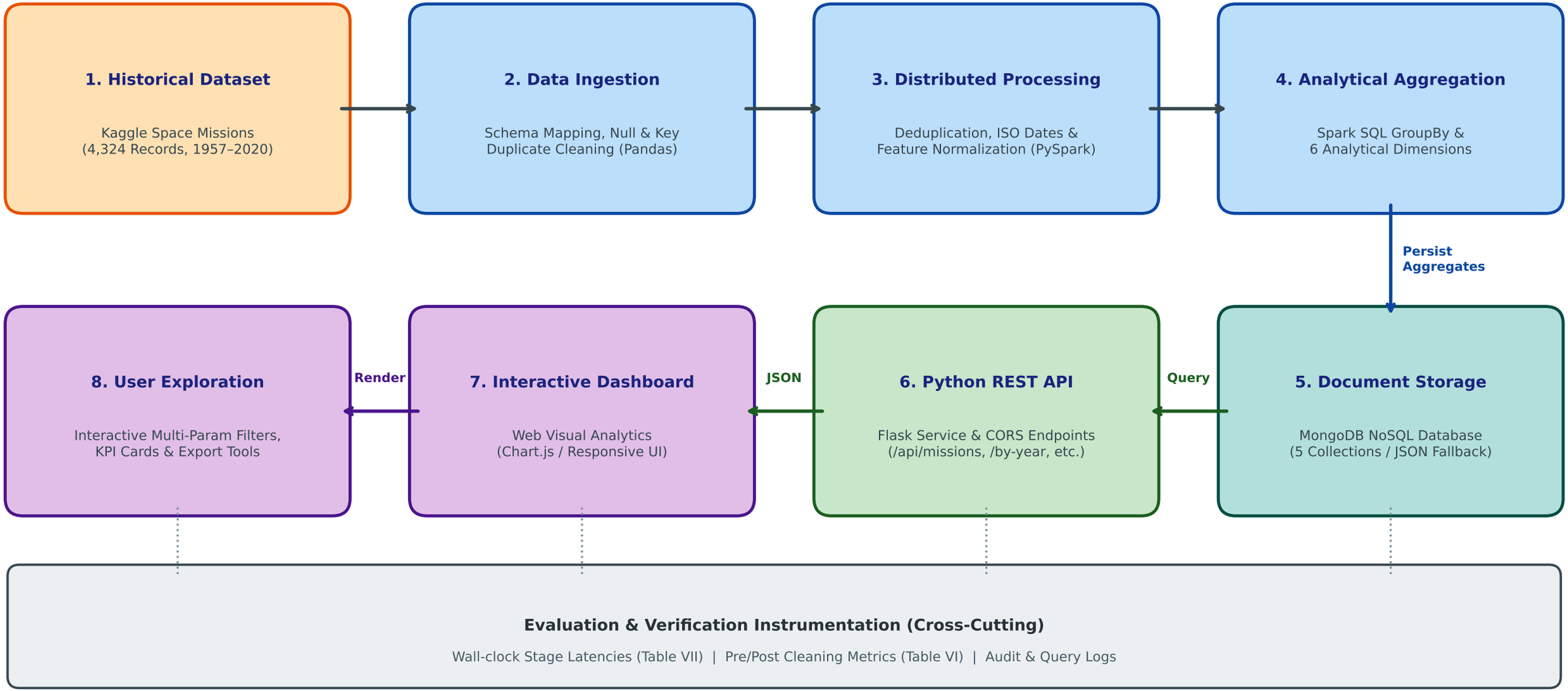
1. Historical Dataset
Kaggle Space Missions
(4,324 Records, 1957–2020)
2. Data Ingestion
Schema Mapping, Null & Key
Duplicate Cleaning (Pandas)
3. Distributed Processing
Deduplication, ISO Dates &
Feature Normalization (PySpark)
4. Analytical Aggregation
Spark SQL GroupBy &
6 Analytical Dimensions
Persist
Aggregates
5. Document Storage
MongoDB NoSQL Database
(5 Collections / JSON Fallback)
Query
6. Python REST API
Flask Service & CORS Endpoints
(/api/missions, /by-year, etc.)
JSON
7. Interactive Dashboard
Web Visual Analytics
(Chart.js / Responsive UI)
Render
8. User Exploration
Interactive Multi-Param Filters,
KPI Cards & Export Tools
Evaluation & Verification Instrumentation (Cross-Cutting)
Wall-clock Stage Latencies (Table VII) | Pre/Post Cleaning Metrics (Table VI) | Audit & Query Logs